

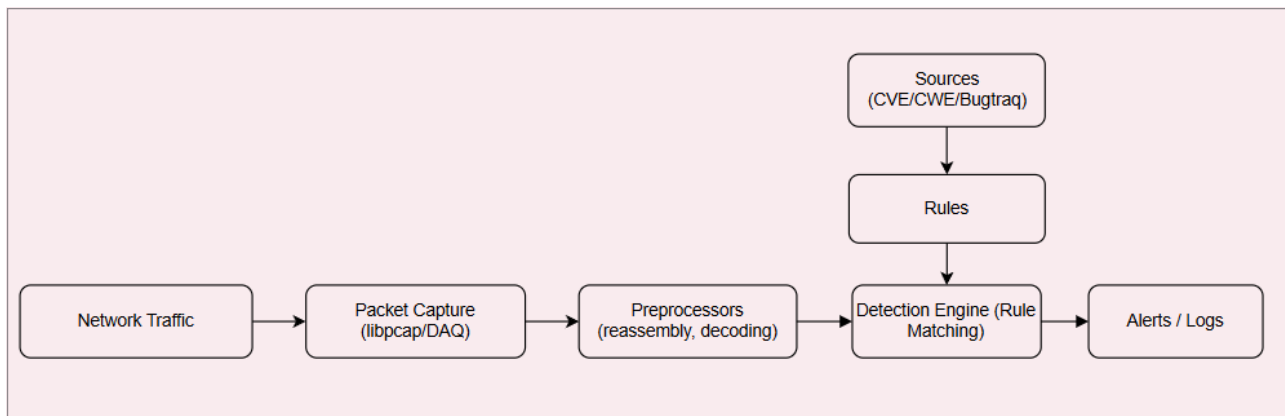


Figure1: Internal working of Snort

system will be ready for the smooth installation of Snort.

Installation of Snort on Ubuntu

Before the installation of Snort, find the IP address and network interfaces of the system using `$ ip addr`; it is necessary to take note of both, since these will be used in the configuration of Snort.

To install Snort and its dependencies use the following command:

```
$ sudo apt-get install snort
```

After successful installation of Snort, check its version using:

```
$ snort --version
```

During installation, Snort automatically creates several directories to store its files, which include configuration files, rules, and logs. The important directories created include:

- `/etc/snort`: This is the main directory for Snort users. It contains configuration files and rule files. The important `.conf` file is also located here, which will be used later for settings.
- `/usr/sbin/snort`: This contains the Snort executable binary, which runs the IDS program.
- `/usr/lib/snort`: This directory holds shared libraries and files used by the Snort executable and other components. These files are

necessary for Snort to work with modules and plugins.

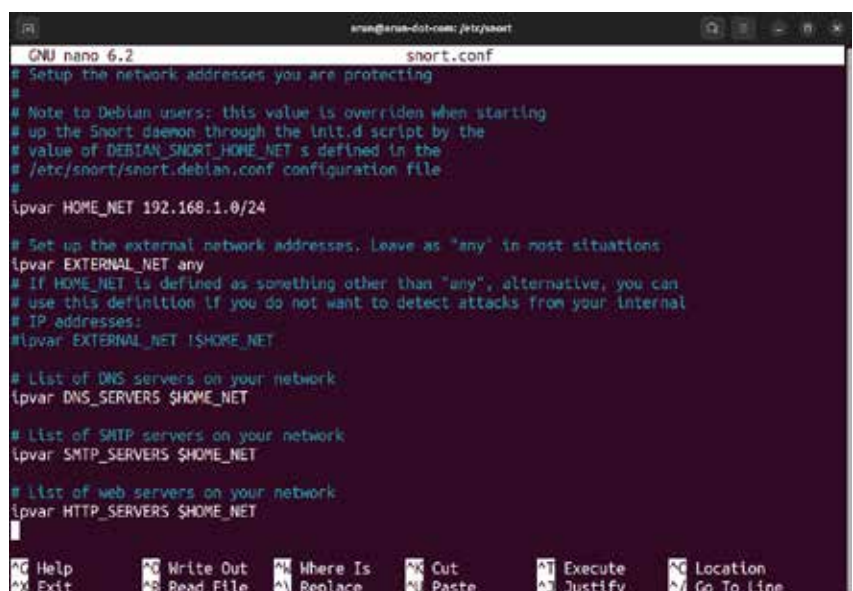
- `/usr/include/snort`: This directory has header files and related resources. These files are used when building custom plugins, preprocessors, or extensions for Snort.
- `/etc/snort/rules`: This directory contains the rule files that Snort uses to detect suspicious activity.
- `/var/log/snort`: This directory keeps the alerts and log files that Snort creates when it finds something.

These directories form the core structure of Snort and will be used in the later steps of configuration. Let's begin with the Snort configuration.

Setting `HOME_NET`

First, navigate to the Snort directory using `$ cd /etc/snort`. Take a backup of the `snort.conf` file using `$ sudo cp snort.conf snort.conf.backup`. Open `snort.conf` file using `$ sudo nano snort.conf`.

The configuration of Snort requires setting the `HOME_NET` variable. The network's IP address and CIDR block (e.g., `192.168.1.0/24`) are first determined using the `$ ip addr` command. Subsequently, the `snort.conf` file is edited, and the default value of the `HOME_NET` variable is replaced with the specific network's IP range as shown in Figure 2. This action allows for the correct identification of internal traffic.

Figure 2: Modifying the `HOME_NET` variable with the IP address found using `$ ip addr` command